Global Video Game Market Trends Analysis (2010–2020)

Sub-title: *Identifying the "Five-Platform Ceiling" through Full-Stack Data Engineering*

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1. Executive Summary

This report outlines an end-to-end data pipeline designed to analyze a decade of video game industry trends. By extracting metadata from the RAWG API and processing it through a relational MySQL environment, this project identifies a critical strategic threshold—the **"Five-Platform Ceiling"**—where multi-platform distribution meets diminishing returns in average sales volume.

2. The Technical Stack

- **Extraction:** Python (Requests, Pandas)
- **Database:** MySQL (Relational Schema Design)
- **Visualization:** Tableau Desktop
- **Systems Rigor:** Applied **A+ Certified** methodology to ensure data integrity and optimized local database performance.

3. Data Architecture & Pipeline

The project followed a strict **ETL (Extract, Transform, Load)** workflow:

1. **Extraction:** Developed a custom script (MiddleMan.py) to interface with the RAWG API, handling JSON parsing and rate-limiting.
2. **Transformation:** Normalized raw API strings into structured data formats, removing outliers and ensuring consistent date/genre categorizations.
3. **Storage:** Engineered a **Relational Database** in MySQL consisting of a **Master Table** (Metadata) and a **Transaction Table** (Sales/Yearly Performance) linked via GameID.

4. Key Business Insight: The "Five-Platform Ceiling"

Analysis of over 10 years of data revealed a distinct peak in Average Units Sold when a title is released on **exactly five platforms**.

- **The Findings:** While expanding from 2 to 5 platforms shows a linear increase in revenue reach, titles released on 6 or more platforms exhibited a **plateau or slight decline** in marginal returns.
- **Strategic Recommendation:** For mid-to-large publishers, resources are most efficiently allocated by targeting a "High-Five" platform strategy. Beyond this point, the increased costs of porting, QA, and multi-platform maintenance often outweigh the additional market share gained.

5. Visual Analysis (Tableau)

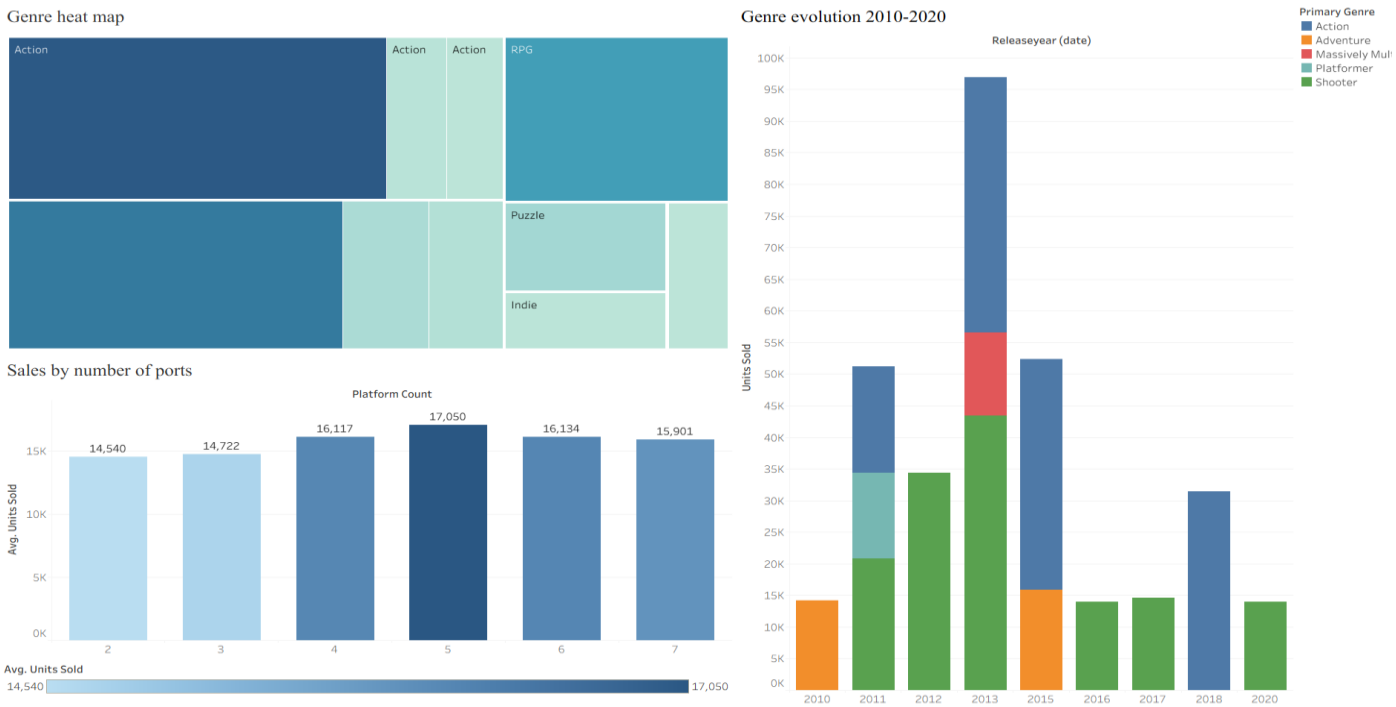

Figure 1: Comparison of Average Units Sold vs. Platform Count, highlighting the ROI peak at the 5-platform mark.

6. Conclusion

This project demonstrates the ability to transform raw, unstructured API data into high-level business intelligence. By leveraging a full-stack technical approach, we moved beyond "tracking sales" to identifying an actionable economic rule-of-thumb for the gaming industry.